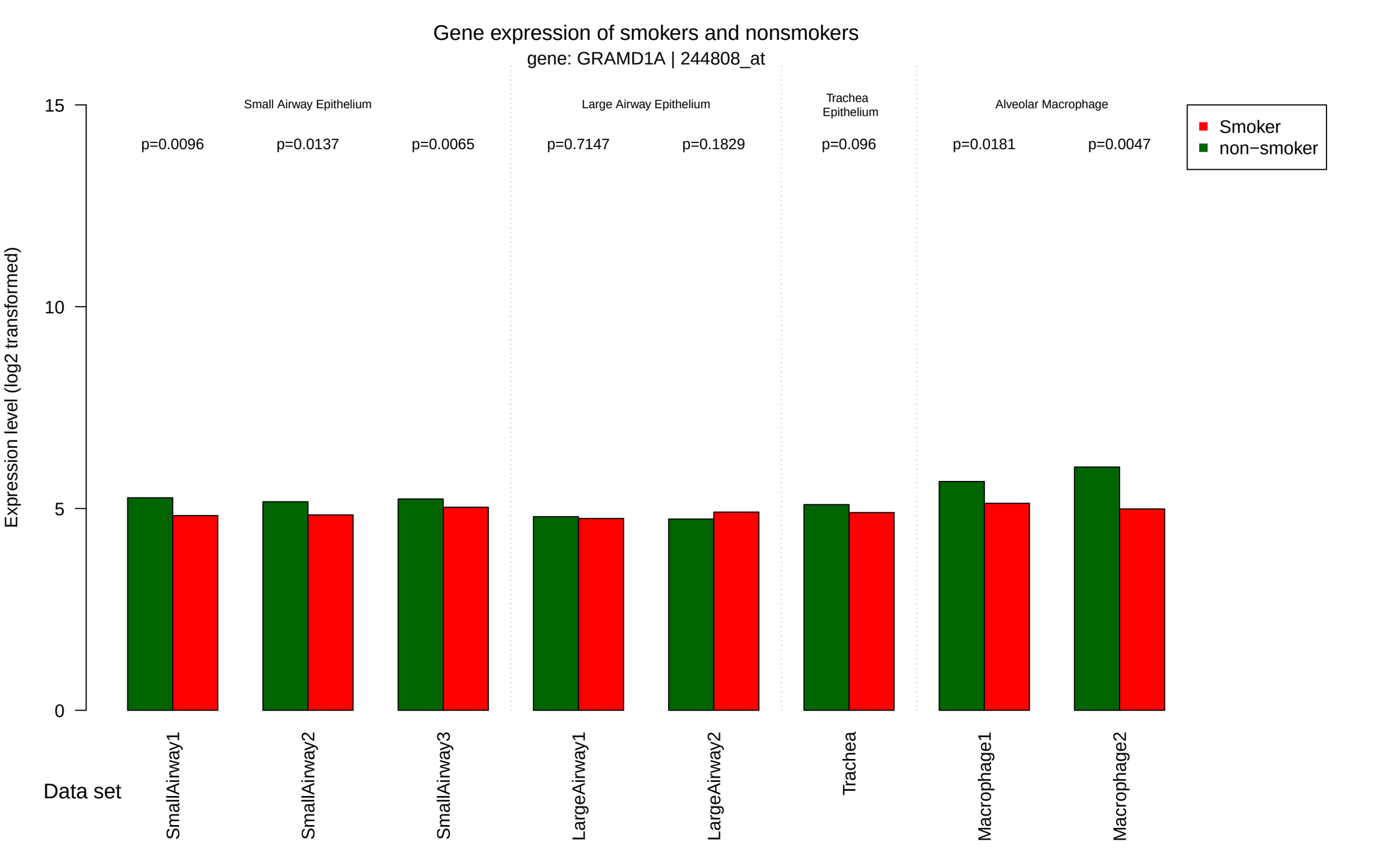
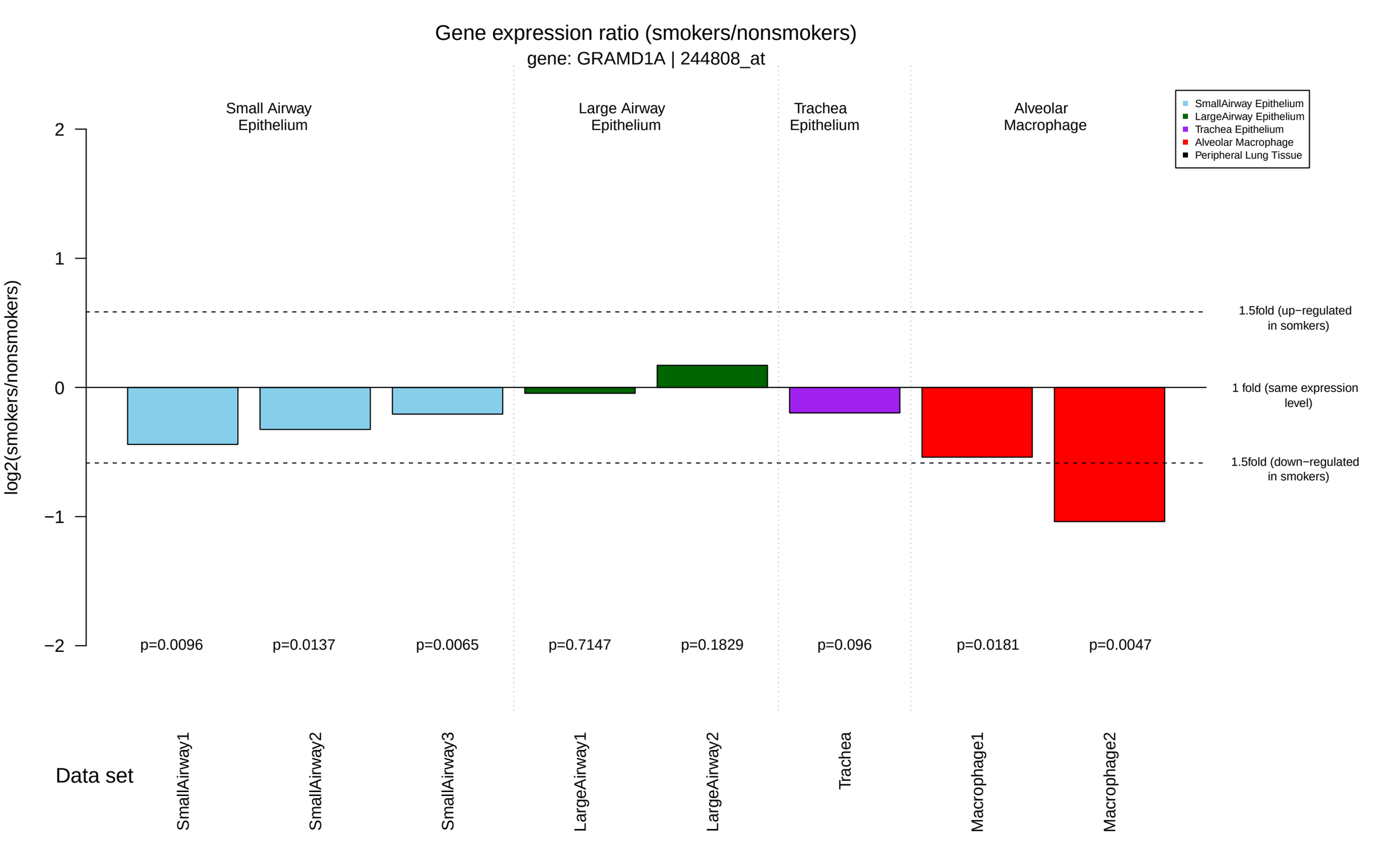
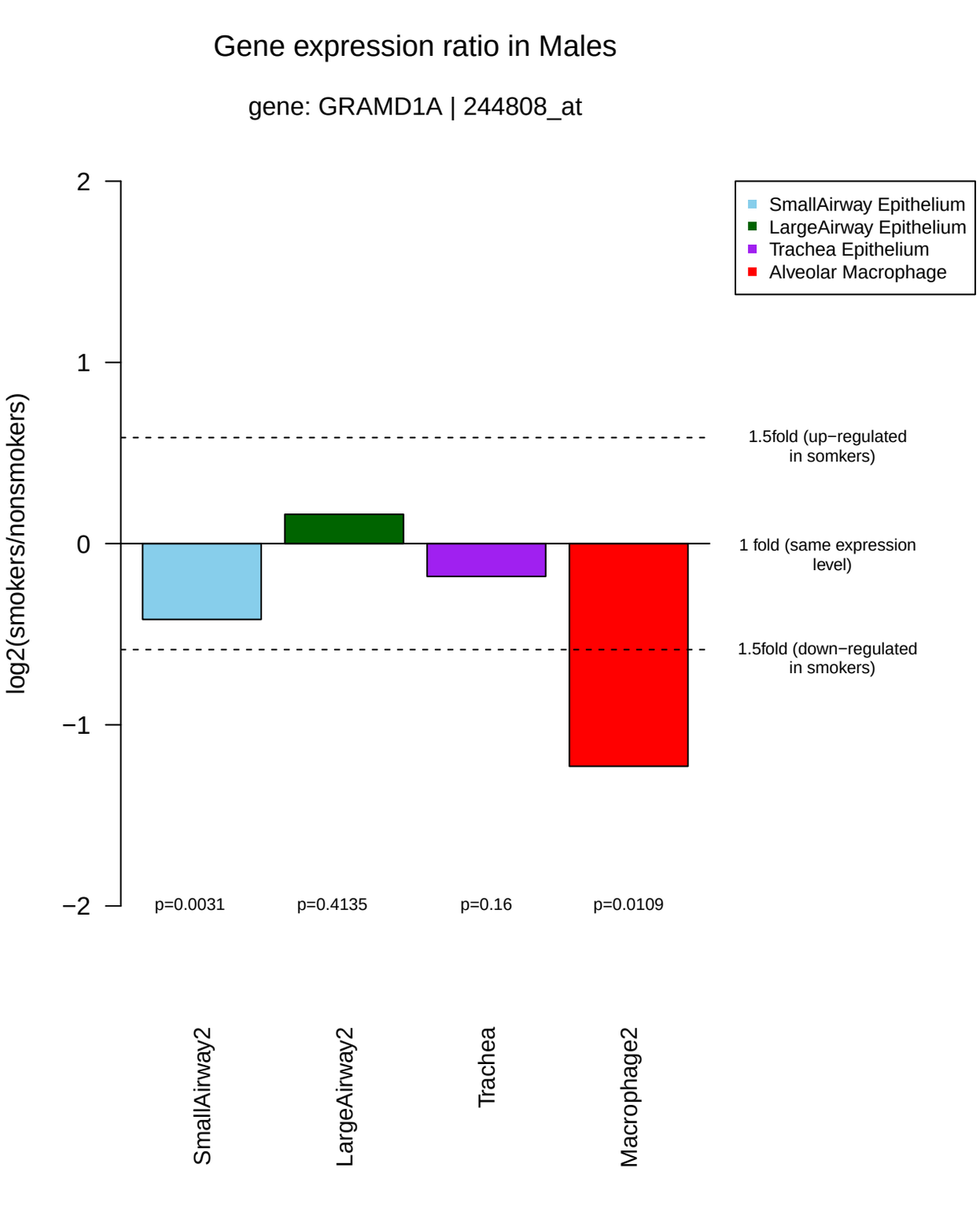
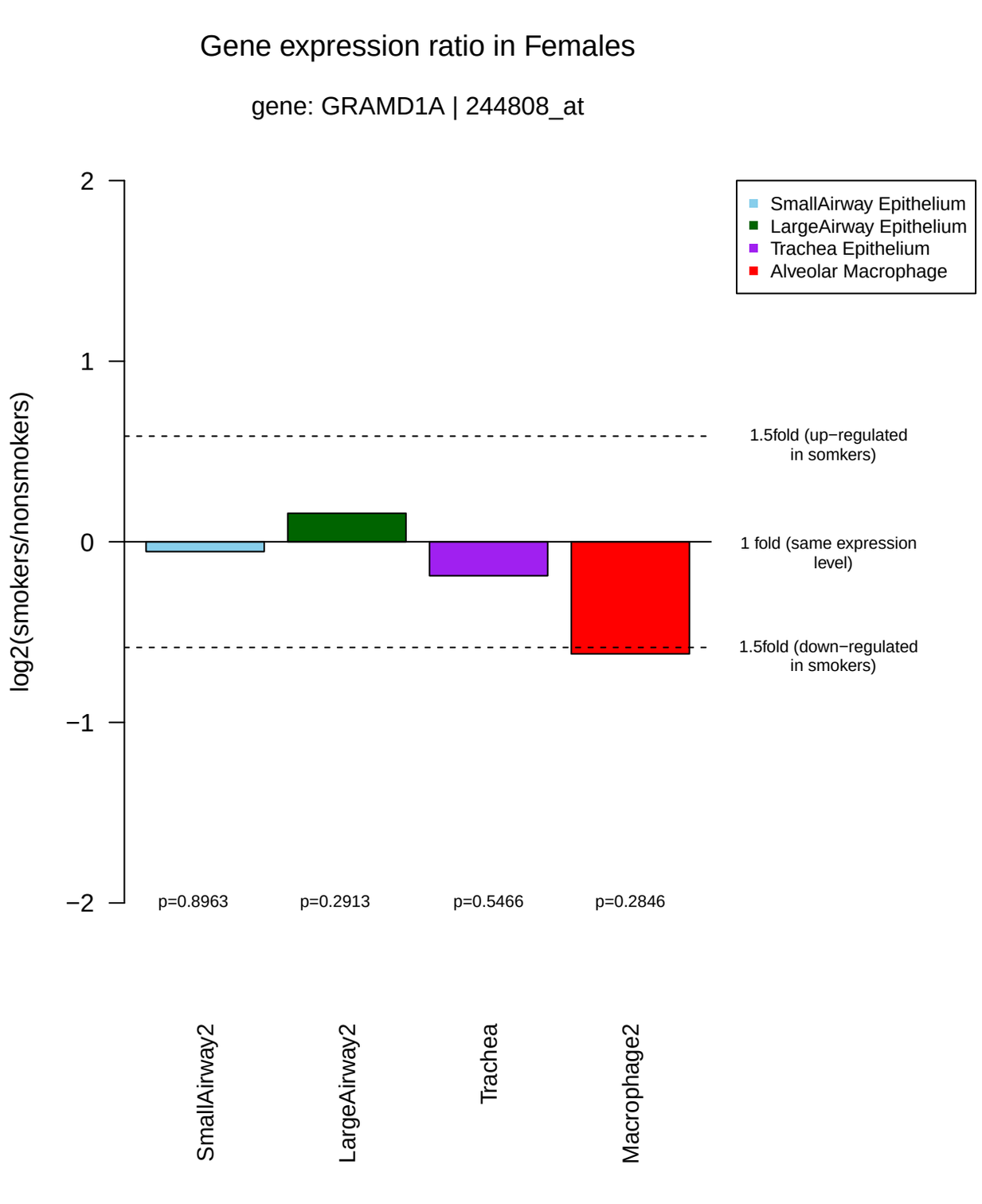
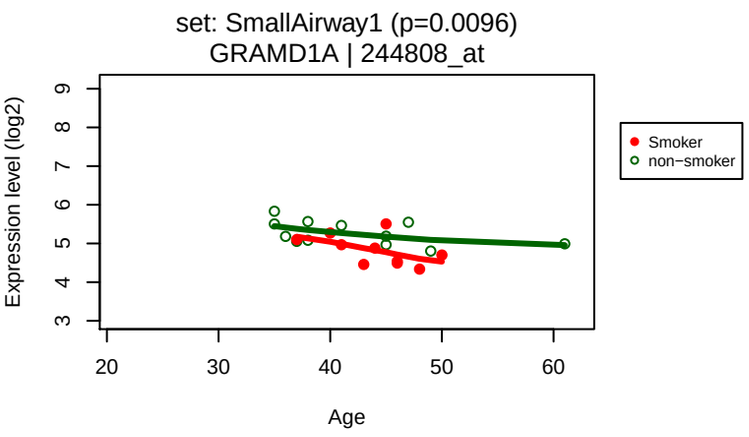
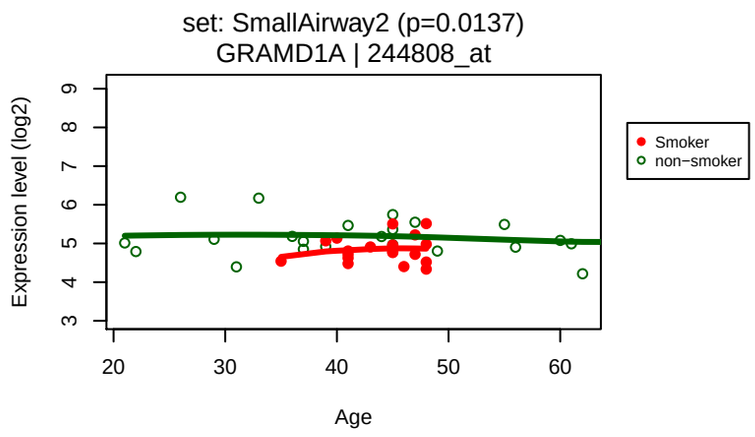

Gene expression of smokers and nonsmokers
gene: GRAMD1A | 244808_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0096
p=0.0137
p=0.0065
p=0.7147
p=0.1829
p=0.096
p=0.0181
p=0.0047
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

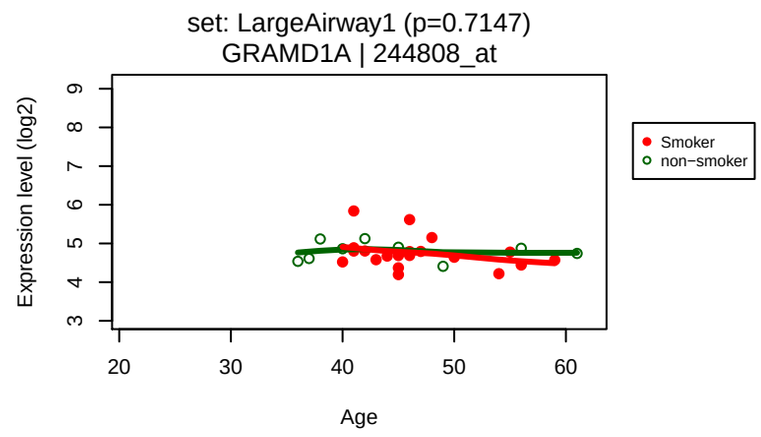

Gene expression ratio (smokers/nonsmokers)
gene: GRAMD1A | 244808_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0096
p=0.0137
p=0.0065
p=0.7147
p=0.1829
p=0.096
p=0.0181
p=0.0047
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

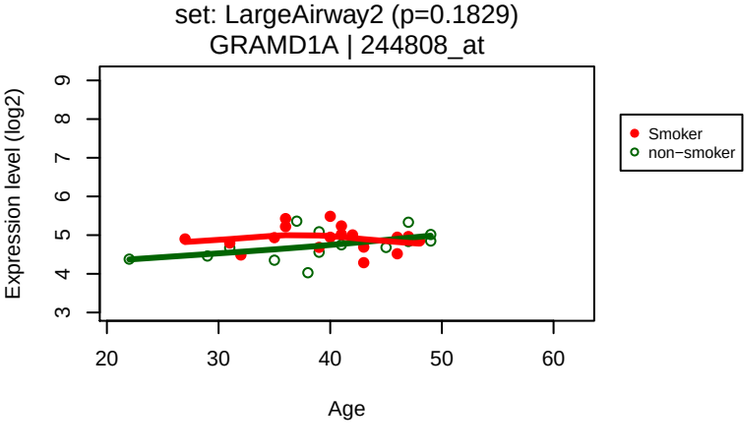

Gene expression ratio in Males
gene: GRAMD1A | 244808_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0031
p=0.4135
p=0.16
p=0.0109
SmallAirway2
LargeAirway2
Trachea
Macrophage2

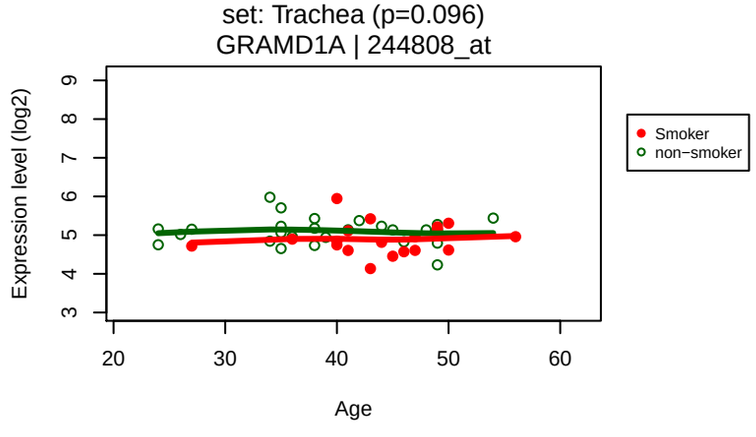

Gene expression ratio in Females
gene: GRAMD1A | 244808_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8963
p=0.2913
p=0.5466
p=0.2846
SmallAirway2
LargeAirway2
Trachea
Macrophage2

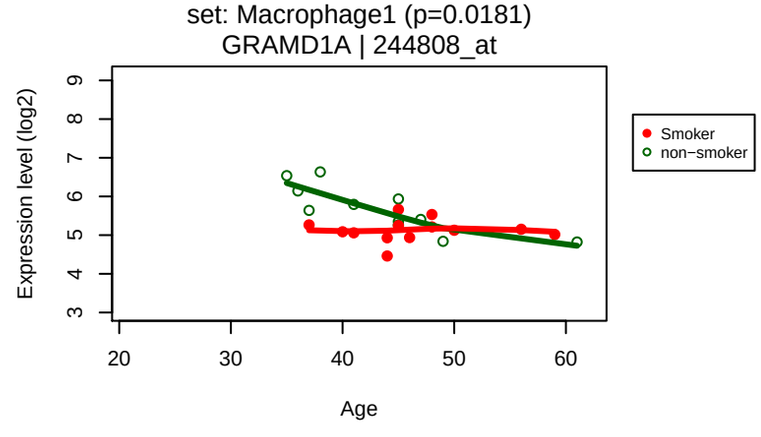

set: SmallAirway1 (p=0.0096)
GRAMD1A | 244808_at
Expression level (log2)
Age
Smoker
non−smoker

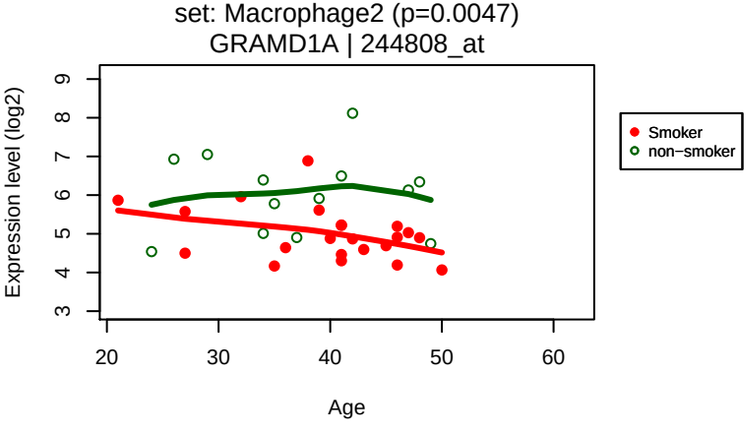

set: SmallAirway2 (p=0.0137)
GRAMD1A | 244808_at
Expression level (log2)
Age
Smoker
non−smoker

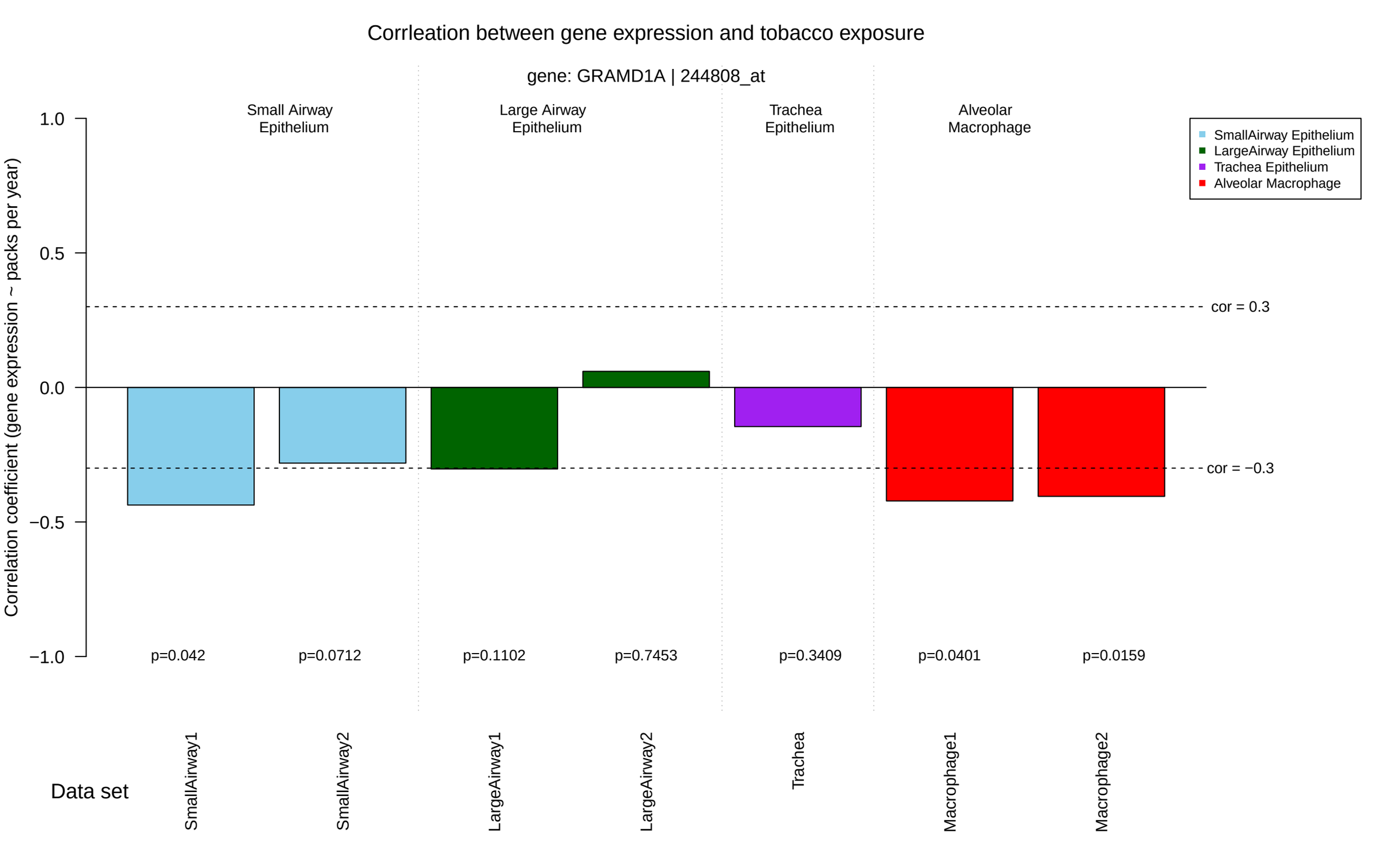

set: LargeAirway1 (p=0.7147)
GRAMD1A | 244808_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.1829)
GRAMD1A | 244808_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.096)
GRAMD1A | 244808_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0181)
GRAMD1A | 244808_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0047)
GRAMD1A | 244808_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: GRAMD1A | 244808_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
Correlation coefficient (gene expression ~ packs per year)
p=0.042
p=0.0712
p=0.1102
p=0.7453
p=0.3409
p=0.0401
p=0.0159
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2